

Tools Integration in Jenkins: GIT & MAVEN

To change the URL of Jenkins Server:

Dashboard → Manage Jenkins → System Configuration → click on System → Jenkins URL → make changes to URL here → Apply + Save

Integration of GIT:

First of all, install GIT on our master server (on Ec2 machine where Jenkins is installed). When you want to integrate any tool and installing plugin of that tool in Jenkins that tool has to be first installed on master Server, plugin installation doesn't mean tool has been installed on master server.

To clone a public GIT Repo:

New Item → add job name → Freestyle Project → OK → select Git → Add Repository URL → add branches to build → Apply + Save → Build Now

If you make any changes to git repository and build again those changes you can be seen in Build with commit id.

To clone a Private GIT Repo:

We can set our gits personal access tokens in Jenkins by using credentials.

First create credentials:

Dashboard → manage Jenkins → Security → Credentials → Click on Global (You can use this for all jobs) → Add Credentials → Select kind - Username with password → Add username → add git hub token for password → add ID → add description → click on Create (You can select - Treat username as a secret - so username will not be visible, you cannot see password again, you can change the password but original password cannot be viewed again)

WhatsApp x Meet - pyq-hkbz-cwa x Instances | EC2 | ap-south-1 x New credentials [Jenkins] x General x + - X

← → ↻ Not secure | 3.6.88.146:8080/jenkins/manage/credentials/store/system/domain/_/newCredentials

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >

New credentials

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

test

☐ Treat username as secret ?

Password ?

....

ID ?

test

Description ?

test

Create

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Search

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Create a new Job:

New Item → give name → Freestyle Project → OK → select Git Plugin → Add Repository URL → add Credentials → add branches to build → Apply + Save → Build Now

Integration of maven:

Install maven on master server first,

Another way of installing maven is through zip:

```
cd /mnt
Mkdir build_tools
Cd build_tools/
Wget https://dlcdn.apache.org/maven/maven-3/3.9.5/binaries/apache-maven-3.9.5-bin.zip
Unzip apache-maven-3.9.5-bin.zip
Rm -rf apache-maven-3.9.5-bin.zip
Cd apache-maven-3.9.5-bin/
Cd bin
./mvn
```

TO set environment variables

```
Cd
Ls -ltra
vi .bash_profile
```

```
# .bash_profile
```

```
# Get the aliases and functions
```

```
if [ -f ~/.bashrc ]; then
```

```
    ~/.bashrc
```

```
fi
```

```
# User specific environment and startup programs
```

```
PATH=$PATH:$HOME/bin
```

```
export MAVEN_HOME=/mnt/build_tools/apache-maven-3.9.5
```

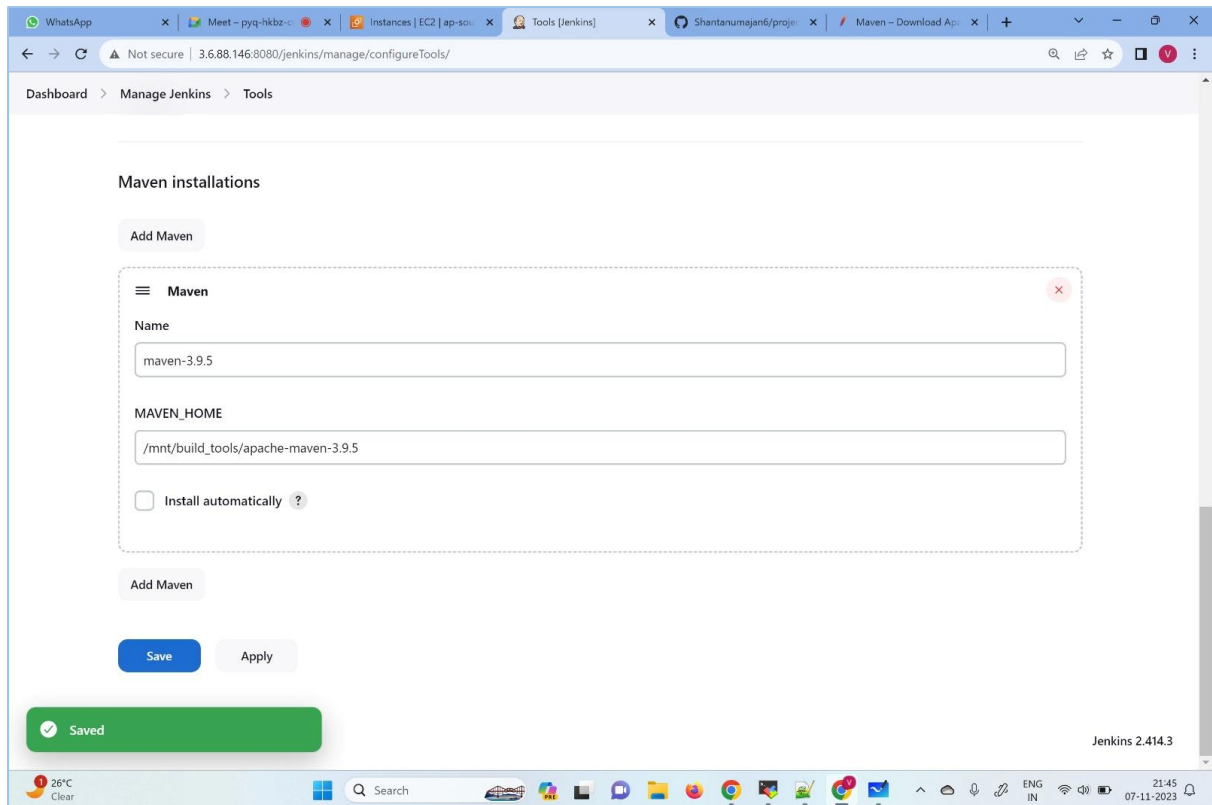
```
export PATH=$MAVEN_HOME/bin:$PATH
```

```
export PATH
```

```
ESC + :x
```

Now logout and login to the root user again and we can see mvn command can run from any location

```
logout
sudo su
```

Dashboard → New Item → give name → freestyle project → discard old builds → use custom workspace → enter directory path /mnt/vel-app → git → add git repository URL (<https://github.com/Shantanumajan6/project>) → select master branch → select delete workspace before the build start → add build steps → execute shell →

```
Rm -rf /root/.m2/repository
```

→ add build step → Import top level maven targets → select maven version → Goals as `clean install` → add build step → Execute Shell →
`cp -r target/LoginWebApp.war /mnt/servers/apache-tomcat-9.0.82/webapps`
 → Apply + Save

Click on Build Now

Hit URL in browser:

public_ip:8080/LoginWebApp

Eg: <http://3.6.88.146:8080/LoginWebApp/>

How to add Plugins in Jenkins:

Dashboard – Manage Jenkins – System Configuration – Plugins – Available plugins – search the name of the plugins – select plugin – click on install

Plugins path on server:

```
cd /root/.jenkins/plugins/
```

Official Website:

plugins.jenkins.io